

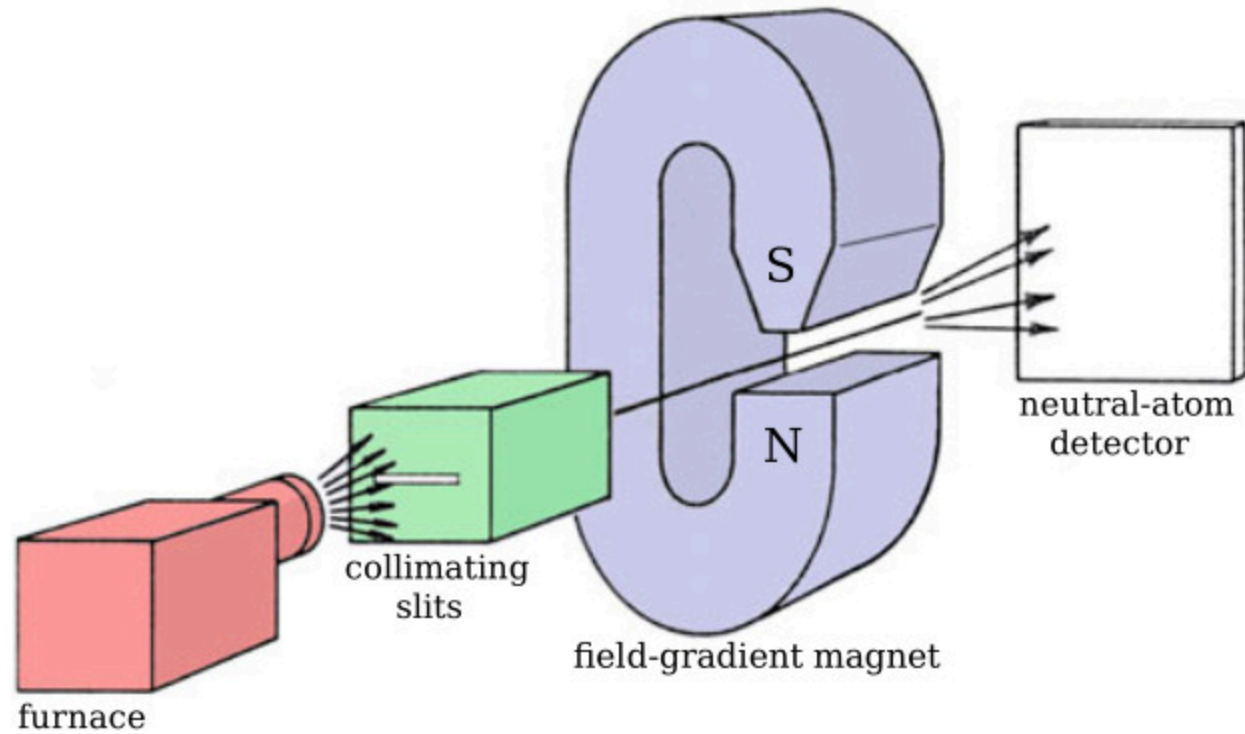
Stern-Gerlach Experiment

Luca Cicu - Lorenzo Liuzzo

27/11/2025

Experiment description

Stern-Gerlach Experiment Apparatus



Angular & Magnetic momenta

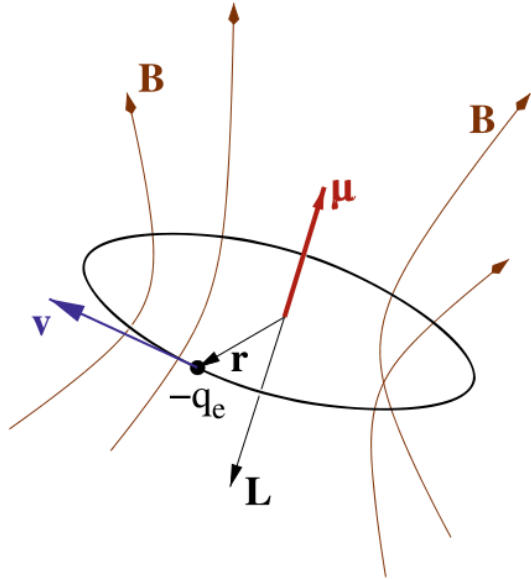


Figure 2: **Angular** and **magnetic** momenta generated by an electron orbiting circularly.

Angular & Magnetic momenta

Angular momentum

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

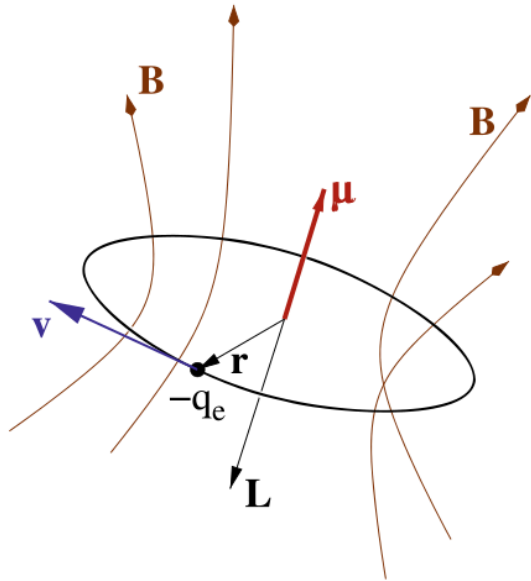
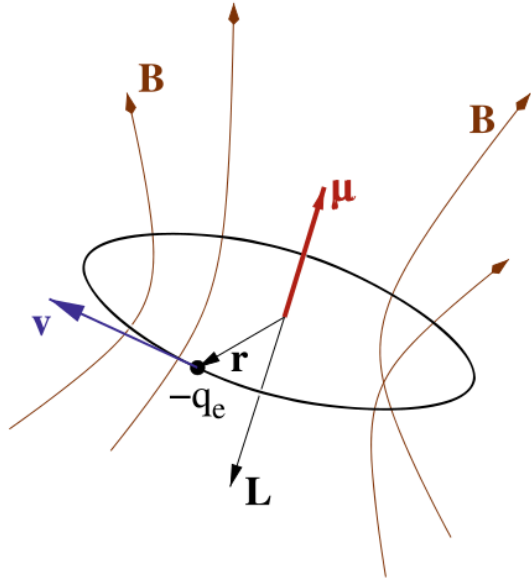


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Angular & Magnetic momenta



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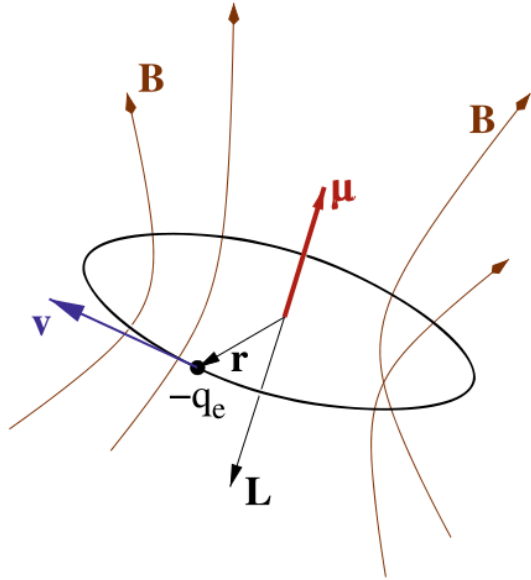
Magnetic momentum

$$\boldsymbol{\mu} = I \Sigma \hat{\mathbf{n}}$$

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Stern-Gerlach Experiment

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$$\boldsymbol{\mu} = \frac{qv}{2\pi r} \pi r^2 \hat{\mathbf{n}} = \frac{q}{2m} \mathbf{L} \Rightarrow \boldsymbol{\mu} \propto \mathbf{L}$$

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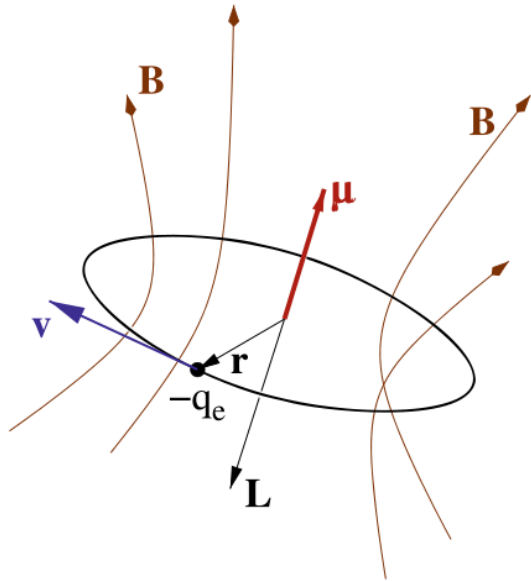


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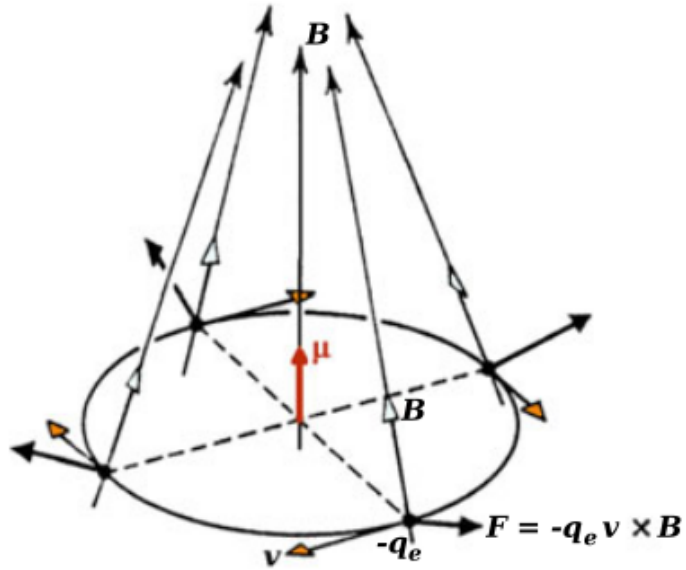
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Magnetic Hamiltonian

$$H_{\text{magn}} = -\boldsymbol{\mu} \cdot \mathbf{B}$$

Stern-Gerlach Experiment

Magnetic Interaction

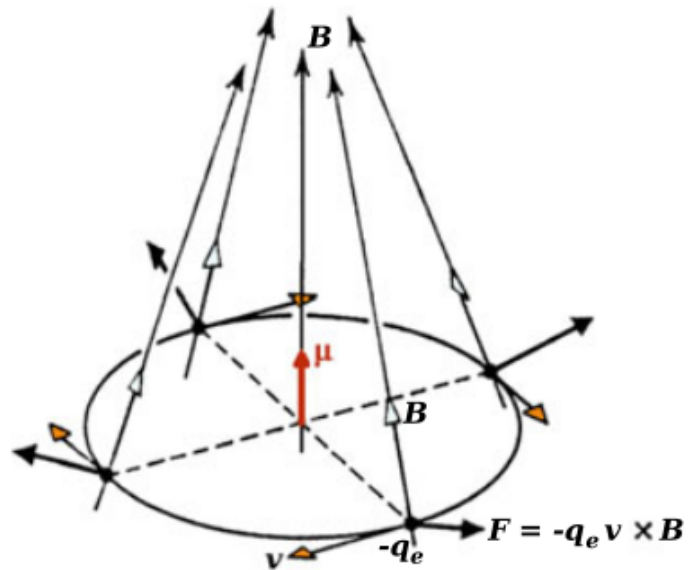


$$F = -\nabla H_{\text{magn}} = \nabla(\mu \cdot B)$$

Figure 3: Microscopic origin of the magnetic force.

Stern-Gerlach Experiment

Magnetic Interaction



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$$F_z = \mu_z \frac{\partial B_z}{\partial z}$$

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Stern-Gerlach Experiment

Magnetic Interaction

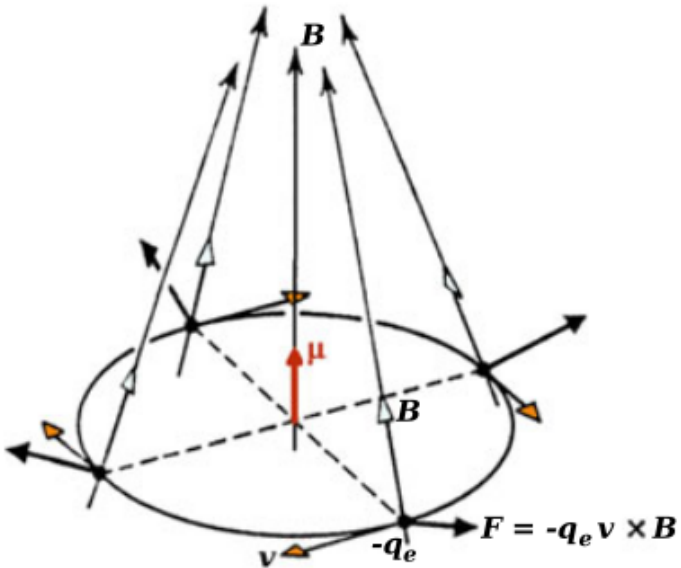


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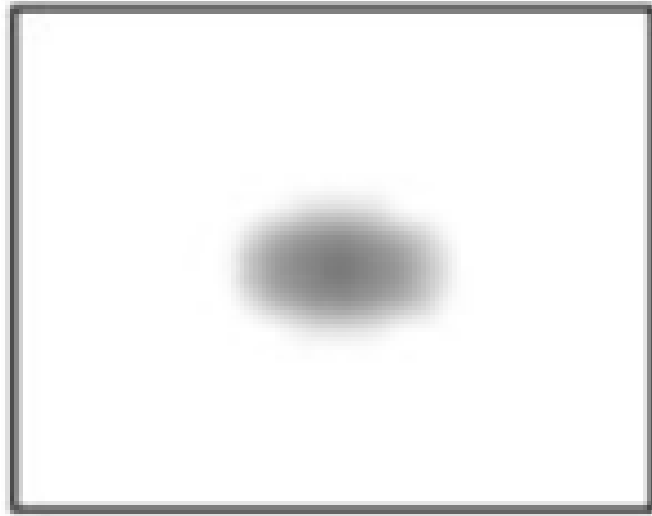
$$F = -\nabla H_{\text{magn}} = \nabla(\mu \cdot B)$$

$$F_z = \mu_z \frac{\partial B_z}{\partial z} = m\ddot{z}$$

$$z(t) = z_0 + \dot{z}_0 t + \frac{\mu_z}{2m} \frac{\partial B_z}{\partial z} t^2$$

Stern-Gerlach Experiment

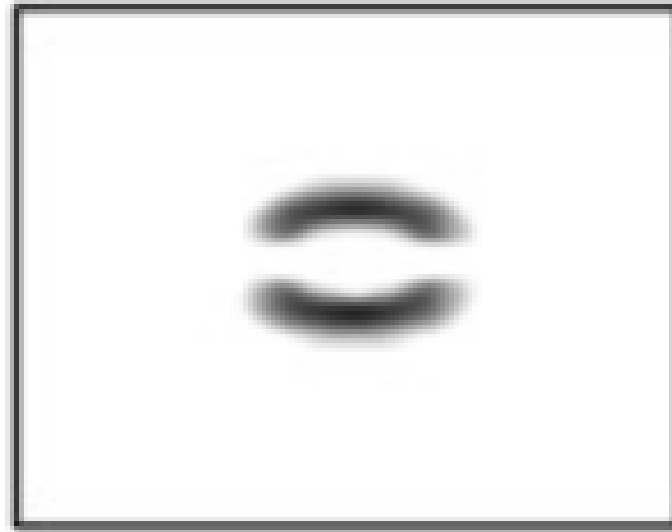
Classical Prediction



classically predicted

Stern-Gerlach Experiment

Observed Results



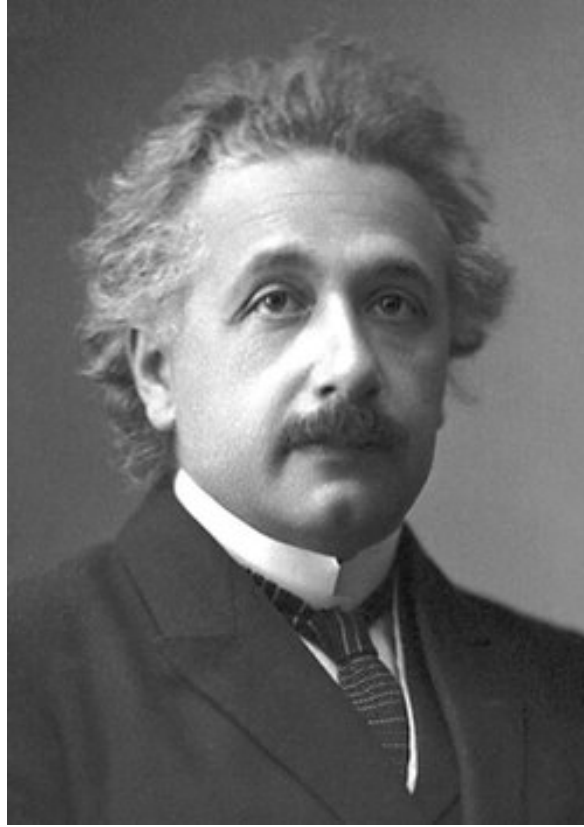
observed

Why two deflected spots?

Historical Context

Stern-Gerlach Experiment

Historical Context



Quantum Mechanics

QM — States in Hilbert Space

$$|\psi\rangle \in \mathbb{H}$$

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- **Basis Properties**

$$\langle e_i | e_j \rangle = \delta_{ij} \quad I = \sum_i |e_i\rangle \langle e_i|$$

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$$|\psi\rangle = \sum_i |e_i\rangle \langle e_i | \psi \rangle = \sum_i c_i^\psi |e_i\rangle$$

$c_i^\psi = \langle e_i | \psi \rangle$ are **probability amplitudes**

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- **Born's rule**

$$\langle \psi | \psi \rangle = \sum_i (c_i^\psi)^* c_i^\psi = \sum_i |c_i^\psi|^2 = 1$$

$|c_i^\psi|^2$ is the **probability** to find $|\psi\rangle$ in $|e_i\rangle$

A **linear operator** $\hat{O}$ is a transformation from an Hilbert space into itself, that acts like

$$\hat{O}(\alpha|\psi\rangle + \beta|\varphi\rangle) = \alpha\hat{O}|\psi\rangle + \beta\hat{O}|\varphi\rangle = \alpha|\psi'\rangle + \beta|\varphi'\rangle$$

Stern-Gerlach Experiment

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Stern-Gerlach Experiment

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We can also define the **adjoint** operator $\hat{O}^\dagger$: $\langle\psi'| = \langle\psi|\hat{O}^\dagger$ $\hat{O}_{ij}^\dagger = (\hat{O}_{ji})^*$

QM — Operators & Observables

We can associate every **observable** of a system to an operator

$$\hat{O} = \sum_i \lambda_i |e_i\rangle \langle e_i| \quad \hat{O}|e_j\rangle = \sum_i \lambda_i |e_i\rangle \langle e_i|e_j\rangle = \lambda_j |e_j\rangle$$

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where λ_i are the **possible results** and $|e_i\rangle$ are the **possible states** in which the system can be founded **after** the measurement.

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Such operators are called **hermitian** and satisfy $\hat{O}_{ij}^\dagger = \lambda_i^* \delta_{ij} = \lambda_i \delta_{ij} = \hat{O}_{ij}$

QM — Commutator of Operators

We define the **commutator** between two operators $\hat{A}$ and $\hat{B}$ as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = -[\hat{B}, \hat{A}]$$

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Two observables can be **simultaneously** determined iff their operators are compatible, that is if they share a **common basis** of eigenstates:

$$\hat{A}|e_i\rangle = \lambda_i|e_i\rangle \quad \hat{B}|e_i\rangle = \mu_i|e_i\rangle \quad \Leftrightarrow \quad \langle e_i|[\hat{A}, \hat{B}]|e_j\rangle = (\lambda_i\mu_i - \lambda_j\mu_j)\langle e_i|e_j\rangle = 0$$

Canonical commutator

$$[\hat{p}, \hat{x}] = -i\hbar$$

QM — Uncertainty Principle

We define the **uncertainty** of an operator $\hat{A}$ in a state $|\psi\rangle$ as its **standard deviation**:

$$\Delta^2 \hat{A}_\psi = \langle \hat{A}^2 \rangle_\psi - (\langle \hat{A} \rangle_\psi)^2 = \langle (\hat{A} - \langle \hat{A} \rangle_\psi)^2 \rangle_\psi$$

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The uncertainty of a pair of operators must satisfy the **uncertainty principle**:

$$\frac{1}{4} |\langle [\hat{A}, \hat{B}] \rangle_\psi|^2 \leq |\langle \hat{A}\hat{B} \rangle_\psi|^2 \leq \langle \hat{A}^2 \rangle_\psi \langle \hat{B}^2 \rangle_\psi$$

Heisenberg's Principle

$$\Delta^2 \hat{x} \Delta^2 \hat{p} \geq \frac{\hbar^2}{4}$$

QM — Angular Momentum

$$\hat{L}_k = (\hat{\mathbf{x}} \times \hat{\mathbf{p}})_k = \varepsilon_{kij} \hat{x}_i \hat{p}_j$$

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Spectrum

$$\hat{L}^2 |lm\rangle = \hbar^2 l(l+1) |lm\rangle \quad l \in \mathbb{N}$$

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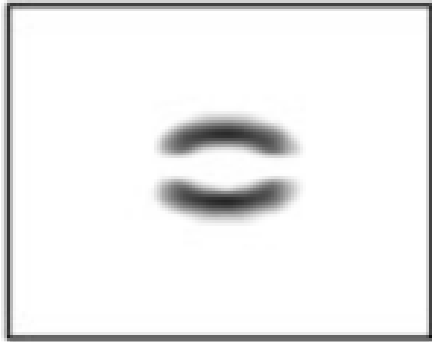
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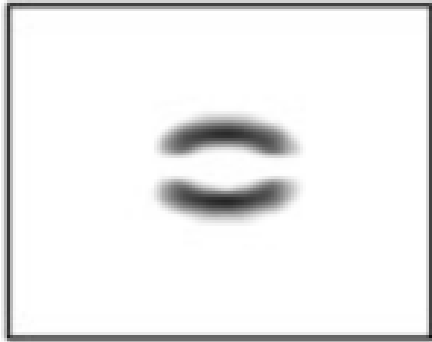
$$\hat{L}_z |lm\rangle = \hbar m |lm\rangle \quad -l \leq m \leq l$$

$$\text{Degeneracy} \quad d_l = 2l + 1$$



observed

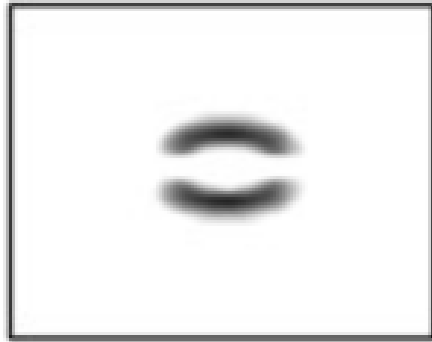
$$\hat{\mu} = \frac{\hbar q_e}{2m_e} \frac{\hat{\mathbf{L}}}{\hbar} = -g_l \mu_B \frac{\hat{\mathbf{L}}}{\hbar}$$



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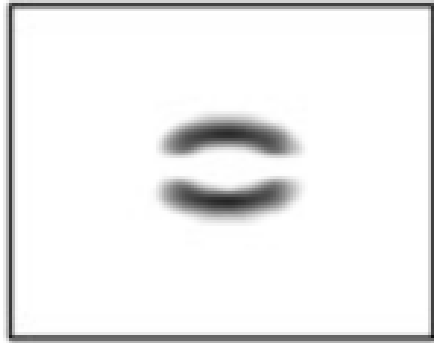


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But $\hat{L}_z$ has $d_l = 2l + 1$ possible eigenvalues, which is **odd**... In the case $l = 0$ we should see just one undeflected spot.

Why two deflected spots?

Spin

QM — Intrinsic Angular Momentum

Commutation rules

$$[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k \quad [\hat{S}_i, \hat{S}^2] = 0$$

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QM — Theoretical Review

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Energy Spectrum

$$E_{m_l m_s} = \frac{\mu_B B_z}{\hbar} (g_l \langle \hat{L}_z \rangle_\psi + g_s \langle \hat{S}_z \rangle_\psi)$$

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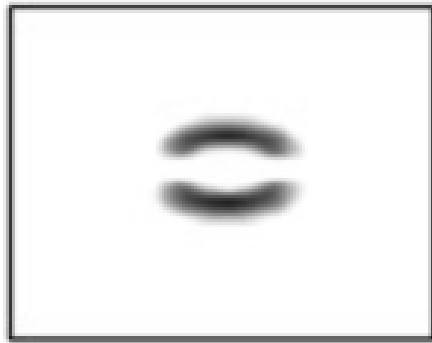
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Degeneracy $d = (2l + 1)(2s + 1)$

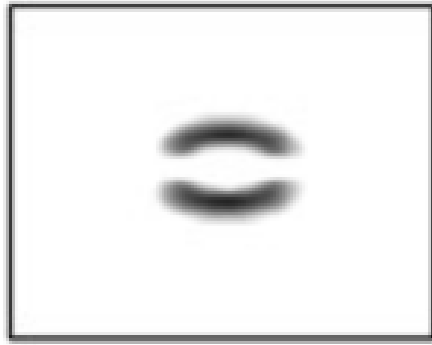
QM — Experimental Outcome

The degeneracy is $d = d_s = 2s + 1 = 2$



observed

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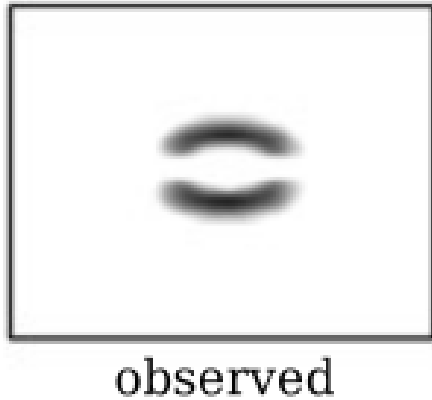


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$$s = \frac{1}{2} \Rightarrow m_s = \pm \frac{1}{2}$$

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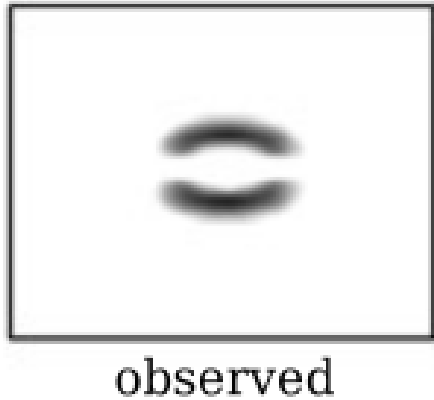


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the SG apparatus is **measuring the spin!**

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the SG apparatus is **measuring the spin!**

By measuring the spacing

$$\Delta E_{0,\pm\frac{1}{2}} = g_s \mu_B B_z$$

we also conclude that $g_s = 2$

Stern-Gerlach Experiment

Conclusion

- Spin can assume both **integer** and **semi-integer** eigenvalues.
- Particles are classified as **bosons** or **fermions**.
- Also nuclei and molecules have a magnetic momentum which can be probed with SG

- *Introduction to the Physics of Matter*, Nicola Manini
 - *Fisica quantistica*, Stefano Forte, Luca Rottoli

Thank you for the attention!

Total Angular Momentum

Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$

Definition

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$$

$$\hat{\mathbf{J}}^2 = \hat{\mathbf{L}}^2 + \hat{\mathbf{S}}^2 + 2\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}$$

Commutation rules

$$[\hat{L}_i, \hat{S}_j] = 0 \quad [\hat{J}_i, \hat{\mathbf{J}}^2] = 0$$

Spectrum

$$\hat{\mathbf{J}}^2 |jm_j\rangle = \hbar^2 j(j+1) |jm_j\rangle \quad |l-s| \leq j \leq l+s$$

$$\hat{J}_z |jm_j\rangle = \hbar m_j |jm_j\rangle \quad -j \leq m_j \leq j$$

Degeneracy

$$d_j = 2j + 1 = (2l + 1)(2s + 1)$$

Coupling of Angular Momenta

Consider the magnetic field experienced by an e^- orbiting around a nucleus with $Z p^+$

$$\mathbf{B}(\mathbf{r}) = \frac{Zq_e}{4\pi\epsilon_0 c^2 m_e} \frac{\hat{\mathbf{L}}}{|\mathbf{r}|^3}$$

Spin-Orbit coupling $\hat{H}_{\text{s-o}} = -\frac{1}{2}\hat{\boldsymbol{\mu}}_s \cdot \mathbf{B}$

$$\begin{aligned}\hat{H}_{\text{s-o}} &\propto \frac{1}{r^3} \hat{\mathbf{S}} \cdot \hat{\mathbf{L}} \\ &\propto \frac{1}{r^3} (\hat{\mathbf{J}}^2 - \hat{\mathbf{L}}^2 - \hat{\mathbf{S}}^2)\end{aligned}$$

We can represent the e^- state using the basis

$$\{\hat{\mathbf{L}}^2, \hat{L}_z, \hat{\mathbf{S}}^2, \hat{S}_z\}$$

Another option is the **coupled** basis

$$\{\hat{\mathbf{J}}^2, \hat{J}_z, \hat{\mathbf{L}}^2, \hat{\mathbf{S}}^2\}$$

since

$$[\hat{J}_z, \hat{\mathbf{L}}^2] = [\hat{J}_z, \hat{\mathbf{S}}^2] = 0$$

$$[\hat{\mathbf{J}}^2, \hat{\mathbf{L}}^2] = [\hat{\mathbf{J}}^2, \hat{\mathbf{S}}^2] = 0$$

Energy Spectra in Magnetic Field

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{s-o}} + \frac{\mu_B B_z}{\hbar} (\hat{L}_z + 2\hat{S}_z)$$

Energy Spectra in Magnetic Field

Paschen-Back Effect $\mu_B B \gg \xi$

$$\xi_{m_l m_s} = \mu_B B_z (m_l + 2m_s)$$

Features

- Spin-orbit coupling negligible
- L and S precess independently around B
- Good quantum numbers: m_l, m_s
- Linear splitting in B

Zeeman Effect $\mu_B B \ll \xi$

$$\xi_{m_j} \cong g_j m_j \mu_B B_z$$

Anomalous Zeeman Effect

- Spin-orbit coupling dominant
- L and S coupled to J
- Good quantum numbers: j, m_j
- Energy shift: $\Delta E = \mu_B B g_j m_j$

Energy Spectra in Magnetic Field

Landé g-factor

$$g_j = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}$$

Stern-Gerlach Experiment

Boson & Fermions

Hello world